Achraf Mzyiene

Railway V&V Engineer | Signalling V&V | ATP/ATO | Train Control Systems

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Professional Summary

Railway V&V Engineer specialised in signalling and train control systems, with 3+ years of experience in embedded and safety-critical systems, including 2 years in railway signalling V&V on ALSTOM projects. Experienced in ERTMS/ETCS, ATP/ATO, requirements verification, functional test design and execution, requirements traceability and CENELEC EN 50126/50128/50129.

Skills

Railway Systems and Signalling: ERTMS/ETCS, ATP/ATO, Train Control, TCMS, DMI, Signalling

Functional Areas: Movement Authority (MA), Speed Supervision, Emergency Braking (EB), Driving Modes, ETCS Level Transitions

V&V: Requirements Traceability, Functional Verification, Requirements Verification, Specification Review, Test Case Design, Test Planning, Test Reporting, Defect Management, Technical Reviews

Standards: CENELEC EN 50126, EN 50128, EN 50129, ISO 26262, MISRA C

Tools: IBM DOORS, ClearQuest, Doc4A, AEC, CANalyzer, MATLAB/Simulink, Git, GitHub, Microsoft Excel (VBA), Word, Visio

Programming and Protocols: Python, C, C++, CAN, LIN, GSM-R

Professional Experience

V&V Engineer – Railway Signalling and Train Control Systems (ATP/ATO)

Vulcain Engineering Group (client: ALSTOM) | Fès-Meknès, Morocco (Hybrid) | 02/2026 – Present

- Review software functional requirements for ATP/ATO functions to ensure coverage, consistency and traceability.
- Develop and update test cases based on functional and safety requirements for ATP/ATO systems.
- Define and execute verification scenarios for ATP/ATO and train control functions: Movement Authority (MA) management, speed supervision, Emergency Braking (EB), and ATP/ATO–TCMS interaction for traction and braking commands.
- Report, track and manage specification defects and anomalies using ClearQuest.
- Participate in design and verification reviews, address technical comments and prepare verification and test reports.
- Collaborate with Design and RAMS teams to resolve technical comments, clarify functional behaviour and ensure consistency between requirements, design and verification.

V&V Engineer – Railway Signalling (ERTMS/ETCS Onboard)

Alten Maroc (client: ALSTOM) | Rabat, Morocco | 10/2024 – 01/2026

- Performed verification and validation of ERTMS/ETCS Onboard signalling functions in accordance with CENELEC EN 50126, EN 50128 and EN 50129.
- Reviewed system and subsystem specifications for clarity and compliance with safety and operational requirements.
- Authored test plans aligned with project requirements and safety objectives.
- Designed and executed functional test scenarios: Movement Authority (MA) management, speed supervision, driving mode and ETCS level transitions, Emergency Braking, DMI application data validation.
- Managed requirements traceability and imported verification results into IBM DOORS to support audit readiness.
- Worked with engineering, quality and safety teams to achieve successful reviews and validations.

Model-Based Design Software Architect (Automotive)

Alten Maroc | Rabat, Morocco | 03/2023 – 10/2024

- Designed AUTOSAR-compliant software components (interfaces, I/O) for the Central ECU (VSM), EE 2020 (NEA) architecture, covering Cluster, Chassis and Braking interfaces; modelled in MATLAB/Simulink (Model-Based Design) from customer requirements.
- Applied MISRA C, MAAB and ISO 26262; debugged with CANalyzer, MATLAB/Simulink and C code; generated code with Embedded Coder.
- Managed versioning on GitHub; collaborated with teams in France, Italy, Romania, China and India.

Education

Engineering Degree in Intelligent Systems | Faculty of Sciences and Technics, Fez | 2020 – 2023

Diploma of Higher Education in Science and Technology (DEUST) | Faculty of Sciences and Technics, Fez | 2018 – 2020

Certifications and Languages

Certification: ISTQB Certified Tester – Foundation Level (CTFL)

Languages: Arabic (native), French (proficient), English (proficient)